

Designed by Remus Teodorescu 2026

LECTURE 11.1

Vision-Based Navigation

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NEED · RULES · NETWORKS



Cars, warehouse robots, underwater vehicles and humanoid robots all steer by what a camera sees. The lecture sets the classical image-processing pipeline against a convolutional network that steers from one frame, and shows why the network needs the right training frames and a GPU to steer in time. The JetRacer Pro is the scaled-down case.

Learning Objectives

SECTIONS, IN ORDER

1. **The need** — vehicles and robots that steer by camera
2. **Conventional vision** — fixed rules, and where they fail
3. **Deep learning** — a network that steers, its data, its GPU

The example is Ex_11's line follower: a study notebook on drawn frames, where the true position of the line is known, so a classical detector and a network can be scored on the same frames. Obstacles are the next lecture.

READ, AND WHAT EACH ANSWERS

UDL ch. 1	where is deep learning applied?
UDL ch. 10	why do images need convolutions?
UDL ch. 8	what must the training data show?

Recap of L10.2 Electrolysers

1 – 2 · A NETWORK FOR THE FUEL CHANNEL

1 **How can a network learn the steam fraction along the channel of the cell for every cell voltage?**

The network takes the position along the channel and the cell voltage, and is trained over 1.10 to 1.40 V until the species equation holds at points along the channel, with the local current set by the gap between the cell voltage and the local Nernst potential. The steam fraction at the inlet is built into the form of the solution, and the zero slope at the outlet is held by a loss term. After one training, any voltage in the range is read off, not solved again. *Liu ch. 2.2; Ex_10.2 section 5*

2 **Why does the local current make the fuel channel harder to solve, and how is the network's answer checked?**

The flux at the electrode is no longer given: the current at each point follows the local steam fraction, so the equation is nonlinear in it and needs a condition at the outlet, and it can no longer be marched from the inlet. The network takes the nonlinear term into its residual with no change of method. Its answer is scored against a boundary-value solver at a tight tolerance: at 1.29 V within 0.0005 of the steam fraction, where finite volumes on 50 cells are within 0.0042. *Ex_10.2 sections 3 – 5*

3 – 4 · THE OPTIMAL VOLTAGE

3 **How can the optimal cell voltage be found with a network, and what does the gradient save?**

The worth of an hour, the hydrogen made at its price less the electricity used at its price, is computed from the network's steam fraction and current. With the voltage as an input, automatic differentiation gives the derivative of the worth with respect to it, and gradient ascent finds 1.30 V in 200 steps and 0.2 s; a sweep of 61 runs of the reference solver finds the same voltage in 4 s. The prices are assumptions, so the voltage found is conditional on them. *Liu ch. 5; Ex_10.2 section 6*

4 **Why does a network help to optimise the operation of the cell over its lifetime, and what should not be claimed from it?**

Over a lifetime the decision is a trajectory of the voltage and the temperature, because what produces more now ages the cell faster. A solver gives the worth, but its gradient costs one solve per variable per step, or an adjoint built by hand; a trained network gives the gradient with respect to every input by automatic differentiation. Degradation rates are uncertain by an order of magnitude, so a predicted lifetime ranks schedules; it is not a number to promise. *Liu ch. 5*

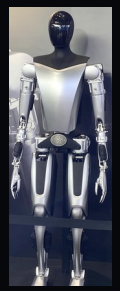
Autonomous Vehicles and Robots

Vision-based navigation — steering a vehicle or a robot from what its camera sees: where the free path is, where it leads, and what is in the way. The camera is cheap and carries the most information of any sensor, but turning its pixels into a decision is the hard part.

THREE OF THEM, AND THE LOOP THEY SHARE



warehouse AGV



humanoid



underwater vehicle

see one frame → **decide** path, obstacles → **act** steer, speed → **in time** fast enough

Photographs: Wikimedia Commons. This lecture is the path; L11.2 is what blocks it.

WHERE THE CAMERA STEERS

	it must see	the hard case
road car	lanes, vehicles, people	night, rain, glare
warehouse robot	aisles, pallets, people	shiny floors, clutter
underwater vehicle	seabed, pipelines, hulls	murky water, little light
humanoid robot	floors, doors, stairs	rooms built for people

Four very different machines, one question: where can I go from here? Under water light is absorbed within metres, red first, which makes the camera's job harder still.

The JetRacer Pro

JetRacer Pro — a small car that computes on board: a Jetson Nano runs the network on its GPU, a wide camera is its only sensor, a servo steers and a brushed motor drives. It is the scaled-down case of the machines on the previous slide, with the same loop: see, decide, act.

TWO CHECKS BEFORE THE FIRST RUN

power mode — the one SETUP.md gives, checked after every boot

the camera mount — move it a centimetre, and every frame is new



the kit, as it arrives

THE VERIFIED SPECIFICATION

compute	Jetson Nano 4 GB, 128-core Maxwell GPU
camera	IMX219-160: 8 MP, 160° field of view
steering	Ackermann, 6 kg·cm servo
drive	4WD, RC380 brushed motor
pack	8.4 V, four 18650 cells, 2S2P
encoder	none

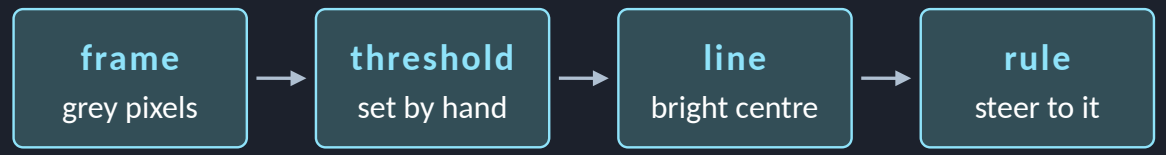
docs/hardware/jetracer-pro.md is the course's checked sheet - quote it, not memory: the early drafts carried the plain JetRacer's pack and drivetrain.

The Classical Pipeline

Conventional vision — fixed image-processing steps chosen by an engineer: the frame is reduced to grey levels, the pixels brighter than the floor are kept, the centre of the line is found, and a hand-written rule turns it into a steering command. Nothing is learned from data.

FOUR STEPS, EACH TUNED BY HAND

FROM A FRAME TO A COMMAND



every step has a number chosen by an engineer, for one floor and one light

It needs no training data, it is fast, and every step can be read and checked. That is why it is still the first thing an engineer tries.

EX_11'S CLASSICAL DETECTOR

the frames	48 × 64 grey, drawn: truth known
the rule	bright pixels in a band of rows
the threshold	0.15 above the floor, by hand
coarsest frame	12 × 16 still meets 0.03
one frame	0.04 ms on a laptop, no training

On the frames it was designed for, the detector is within 0.002 of the true position of the line at full resolution.

Where Conventional Vision Fails

Robustness — keeping the error within what was agreed when the scene changes: the light, a reflection, a shadow, a new floor, an unseen object. A hand-tuned rule is robust only to the changes its designer thought of.

WHAT BREAKS A HAND-TUNED RULE

glare, shadows	bright is not always the line
less light	night, or deep water
a new scene	every threshold tuned again
the unexpected	what no rule was written for

TALK · 2 MIN How would you explain, in your own words, why a rule that works on one floor can fail on another? Then, in the table on the right, what does the detector take for the line in a frame with glare?

THE DETECTOR ON 1000 FRAMES

mean error	detector
along the line	0.024
beside the line	0.026
with glare	0.369
the agreed error	0.03

A patch of glare makes the error fifteen times larger, at every resolution: the glare is bright, and bright is all the rule looks for. A better rule could reject it - once someone has thought of it.

Steering From a Frame

Vision policy — a convolutional network that maps one camera frame to one action. It learns its own features from frames whose true answer is known, instead of following rules written by hand. One output between −1 and +1 makes a steerer; two outputs, free or blocked, detect an obstacle.

EX_11: WHERE IS THE LINE, A LITTLE AHEAD

the input	one 48 × 64 grey frame
the network	three convolutions, two dense layers
the output	the position of the line, one number
the loss	squared error to the true position

MEAN ERROR ON 1000 HELD-OUT FRAMES

	detector	network
along the line	0.024	0.006
beside the line	0.026	0.008
with glare	0.369	0.009
one frame	0.04 ms	0.19 ms
training	none	3 s

The network, shown glare in training, keeps its error below 0.01 where the rule is fifteen times off. The price is training, and five times the time per frame on a laptop CPU - which the next two slides deal with.

Collect the Data Deliberately

Deliberate collection — choosing the frames the policy will be trained on. The network learns only what the pictures show: the floor, the mount and the light it will drive in, and on purpose the positions the car reaches when it is already going wrong, labelled with the way back.

THE RULES, AND THE ONE THAT IS NOT OPTIONAL

where it will drive	same floor, same light, same mount
what disturbs it	glare, shadows, a dirty floor
vary the rest	positions, headings, distances
recovery states	half off the line, at an angle

A policy shown only perfect driving has never seen a mistake, so it cannot correct one: small errors add up, lap after lap.

THE SAME NETWORK, TRAINED TWO WAYS

mean error	along only	deliberate
along the line	0.002	0.006
beside the line	0.136	0.008
with glare	0.149	0.009

Trained only along the line the network is seventeen times worse exactly where a drifting car needs it. Same network, same number of frames, same training time: only the collection differs.

Why a GPU

GPU — a processor with many small cores that run the same operation on different data at the same time. Every output of a convolution is its own sum of products over a small window, independent of the others, so a GPU computes them all at once.

THE DISTANCE DRIVEN WITHOUT NEW INFORMATION

$$d = \frac{v}{f}$$

d the blind distance, v the car's speed, f decisions per second : the whole loop, not the network alone

ONE CONVOLUTION, ALL OUTPUTS TOGETHER



one core per output

the frame, one 3 × 3 window every output at once

TALK · 2 MIN How would you explain, in your own words, why a slightly more accurate but slower network can be a worse driver? Then, in the drawing above, what does the GPU do with the cells on the right?

THE WORK, AND WHAT IT BUYS

Ex_11's network	0.65 million multiply-adds a frame
ResNet-18	1.8 billion, at 224 × 224
Nano GPU	472 GFLOPS at its peak
blind to 10 cm	the car holds the line, 2 cm off
blind at 17 cm	it leaves the course

1 m/s at 10 Hz drives like 2 m/s at 20 Hz: only the blind distance matters. The peak is a ceiling; the real loop rate is measured on the car, capture to command.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
semantic segmentation	which pixels are free to drive on	UDL ch. 10	floors with no line
depth from one camera	distance from a single frame	UDL ch. 10	no range sensor on board
vision transformers	attention over patches of the frame	UDL ch. 12	large, varied data sets
imitation learning	a policy from human driving	UDL ch. 19	road vehicles
reinforcement learning	a policy from reward, by trial	UDL ch. 19	simulators, legged robots

WHY THEY ARE LEFT OUT

Ex_11 follows one taped line with one camera, scored on drawn frames where the true position is known, so none of them is needed to do it. They matter for the full-size machines of the first section.

Exercise 11

ONE STUDY NOTEBOOK, FROM A TO B

- **1 – 3** — the task, the drawn frames, the classical detector
- **4** — a network that steers, trained deliberately
- **5 – 6** — obstacles, and the strategies to avoid them
- **7** — the loop rate, the GPU and fewer bits
- **8 – 10** — the report, and the mini projects



the same task on the real car: a demonstration in the lab

THE QUESTION EACH PART ANSWERS

1 – 3	when is a fixed rule enough?
4	what does deliberate data buy?
5 – 6	which strategy reaches B first?
7	how fast must the loop be?

REPORT QUESTION, TO CONCLUDE

Which avoidance strategy reaches B fastest without a collision, and what limits its speed: the errors of the networks, the loop rate, or the distance needed to react?

SUMMARY

Key Takeaways

VISION POLICY

what it is	a CNN from a frame to an action
function	where the line is, a little ahead
train, tune	frames that show the mistakes
vs a fixed rule	no hand tuning, holds under glare
used for	steering cars, robots, vehicles

THE LOOP ON A GPU

what it is	many cores, one operation each
function	all outputs of a convolution at once
train, tune	the whole loop, measured on the car
vs accuracy	judged together, rate first
used for	a short blind distance $d = v/f$

Next: what blocks the path - obstacles, and what to do about them.

Questions

1 – 2 · A NETWORK THAT STEERS

1 **Why can a convolutional network steer from a camera frame where a hand-tuned line detector fails?**

The detector follows fixed steps chosen by an engineer: it keeps the pixels brighter than the floor and steers to their centre, so anything bright is taken for the line. The network learns its own features from frames whose true position of the line is known. On clean frames both meet the agreed error; with a patch of glare the detector's error rises to 0.37, while the network, shown glare in training, keeps its error below 0.01. *UDL ch. 10; Ex_11 sections 3 – 4*

2 **What must the training frames show for the network to steer a car that is drifting off the line?**

The network learns only what the frames show: the floor, the camera mount and the light it will drive in, the disturbances such as glare, and on purpose the positions the car reaches when it is already going wrong, labelled with the way back. Trained only along the line, the same network is seventeen times worse beside it, which is exactly where a drifting car needs it. Same network, same number of frames: only the collection differs. *UDL ch. 8; Ex_11 section 4*

3 – 4 · THE LOOP ON A GPU

3 **Why is a steering network judged by the rate of the control loop as well as by its error?**

Between two decisions the car drives on its last command with no new information, over a blind distance of its speed divided by the decisions per second. On the simulated lap the car holds the line within 2 cm up to 10 cm blind and leaves the course at 17 cm, so a slightly more accurate network that halves the loop rate is usually a worse car. The whole loop is timed, from capture to command, because the network is only one of its stages. *Ex_11 section 7*

4 **Why does a convolutional network need a GPU to steer in time, and what must be measured on the car?**

Every output of a convolution is its own sum of products over a small window of the frame, independent of the others, so the many cores of a GPU compute them at the same time. A ResNet-18 needs about 1.8 billion multiply-adds for one 224 by 224 frame. The peak rate of the GPU is only a ceiling: the loop rate that sets the blind distance is measured on the car itself, from the frame being captured to the steering command. *UDL ch. 10; Ex_11 section 7*

NEXT

L11.2 — Obstacle Avoidance

What blocks the path: a network that tells free from blocked, the strategies that steer around an obstacle, and the distance the car needs to react.

BEFORE THE NEXT SESSION

read	UDL ch. 10
run	Ex_11's notebook, sections 1 to 4
bring	your detector and network errors
think	what in your frames is an obstacle?

The real JetRacer drives the same task in a lab demonstration, with the throttle limited in software and one hand on the stop.